

MOHAN RAJ G

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github.com/mskmohan50-profile | mskmohan50-profile.github.io/portfolio | Immediate availability

PROFESSIONAL SUMMARY

- Computer Science graduate and Software Developer proficient in React.js, Node.js, Express.js, TypeScript, Python (Flask, FastAPI, Django), and RESTful API development.
- Experienced in full-stack web application development, database design (PostgreSQL, MongoDB, Redis), cloud deployment (AWS, Vercel, Render), containerization (Docker), and JWT-based authentication.
- Built and deployed 3+ production-grade applications, including a full-stack quality intelligence dashboard with role-scoped REST APIs and data visualization, and an LLM-integrated pipeline automating end-to-end regulatory report generation-combining strong analytical and problem-solving skills with hands-on AI/ML system design

EDUCATION

Bachelor of Engineering in Computer Science and Engineering CGPA: 7.0/10 , Coimbatore Institute of Technology	2023 – 05/2026
Diploma in Computer Science 95.5% , Sri Rengaswarar Polytechnic College, Namakkal	2020 – 2023
Higher Secondary School 65% , Govt. Boys Higher Secondary School, Trichy	2018 – 2020

CODING PROFILE

- LeetCode — 150+ problems solved
- CodeChef — 200+ problems solved
- GeeksforGeeks — 100+ problems solved

TECHNICAL SKILLS

Languages

Python, JavaScript, Java, SQL, HTML, CSS

Databases & DevOps

MongoDB, PostgreSQL, Redis, Git, Docker, AWS, GitHub Actions (CI/CD)

Frontend / Backend

React.js, Redux Toolkit, TypeScript, Recharts, Tailwind CSS, Node.js, Express.js, Flask, FastAPI, Django, REST APIs, JWT

AI / ML & Tools

LightGBM, OpenCV, YOLOv5, Hugging Face, LangGraph, Postman, Sentence Transformers, Joblib, Figma

CERTIFICATIONS

Full Stack Development

Project Development Cell, Coimbatore-07/2024

Machine Learning

Sri Regeswarar Collage, Namakkal-05/2023

PROJECTS

AI Agent Workflow Builder

AI Agents • Workflow Automation • Groq API • Next.js • TypeScript • Node.js • Express • PostgreSQL • Webhooks • REST APIs

- Engineered an n8n-style visual workflow automation platform with 6 node types (LLM, HTTP, database, conditional, notification, approval).
- Implemented 4 trigger types (manual, webhook, scheduled, database-event) with real-time execution status tracking.
- Designed RBAC using Nhost Auth and Hasura, including organization management and approval-based workflow pause/resume.
- Integrated Groq API for real-time LLM execution on a modular Node.js/Express backend with PostgreSQL and GraphQL.

AI-Powered Customer Complaint Management System

- Built a full-stack QMS app using a 7-node LangGraph AI pipeline with Groq LLMs to auto-extract data, classify risk, detect duplicates, and recommend CAPA from pharma complaint documents (PDF, DOCX, EML, TXT).
- Implemented real-time auto-save and AI chat-driven form editing using React, Redux Toolkit, and TypeScript.
- Built a FastAPI backend with SQLAlchemy ORM supporting SQLite/MySQL/PostgreSQL, exposing REST APIs for complaint CRUD and AI analysis.
- Designed a sequential multi-agent pipeline (extract → risk classify → CAPA recommend → summarize) using a two-model strategy for latency/quality balance.

DevPilot AI-Autonomous AI Software Engineer

- Developed an autonomous multi-agent AI platform that transforms natural-language requirements into full-stack applications (frontend, backend, database) via a 7-stage agentic pipeline (Requirement → Architecture → Code Generation → Test → Debug → Fix → Review).
- Implemented a self-healing test-fix-retest loop, enabling AI agents to autonomously detect failures, perform root-cause debugging, and repair code.
- Integrated Gemini API for multi-agent orchestration across a Node.js AI engine, FastAPI backend, and React/TypeScript frontend.
- Tested and containerized the full multi-service application (frontend, backend, AI engine) using Docker and Docker Compose, ensuring reproducible deployment.

Manufacturing Quality Intelligence Dashboard, React, TypeScript, Express, MongoDB, JWT, Recharts

- Built a full-stack analytics application with JWT authentication, ownership-scoped REST APIs, and CSV/Excel data ingestion (PapaParse, SheetJS).
- Delivered KPI dashboards with trend and defect visualizations (Recharts) and CSV/PDF export (jsPDF).
- Implemented append-only audit logging, data quality scoring, and a documented multi-tenant scaling architecture.